THE TITLE

Cody A. Campen[1], Timothy R. Brick[1,2], and Sy-Miin Chow[1]

[1]Department of Human Development and Family Studies, The Pennsylvania State University

[2]Institute for Computational and Data Sciences, The Pennsylvania State University

Author Note

Cody A. Campen https://orcid.org/0009-0009-8664-5283

Timothy R. Brick https://orcid.org/0000-0002-3339-9279

Sy-Miin Chow https://orcid.org/0000-0003-1938-027X

The author has no conflicts of interest to disclose.

Correspondence concerning this manuscript should be addressed to Cody A. Campen, Department of Human Development and Family Studies, 237 HHD Building, The Pennsylvania State University, University Park, PA 16802, USA. Email: cac7655@psu.edu

Abstract

THE ABSTRACT.

Keywords: KEYWORD, KEYWORD, KEYWORD

THE TITLE

Binge drinking is bad, it leads to a lot of bad things, and some epi stats, which means we're important.

Binge drinking interventions often try to target...something.

But those somethings are often poorly operationalized or not at all, either through...idk...probably self-report. Studies on self-report show (what, i imagine they're pretty bad, but maybe they're actually surprisingly good). Wearable transdermal alcohol sensors have emerged as an alternative to self-report, offering a passive continuous measure of alcohol consumption (Fairbairn et al., 2025; Gunn et al., 2023). It surpasses previous measures of intoxication (eg breathalyzers) by reducing the burden traditionally involved in obtaining a measurement to (blakn) what were the findings of those feasability studies? But although we have all this data, we dont know what to do with it. Raw transdermal alcohol concentration (TAC) signal can be difficult to interpret and translate into meaningful drinking parameters (Gunn et al., 2023; Kianersi et al., 2023).

To bridge the gap between theory and methods—because this phrasing looks good for pamt— , we propose a formal model for binge-drinking, which operationalizes binged drinking by...—some number— parameters characterizing...—the things we characterize—.

Drawing from previous work on binge-drinking we conceptualize binge-drinking as consisting of discrete drinking episodes of varying magnitude separated by a period of non-drinking (Greenfield, 2000). For example, TAC-derived features such as rise rate, peak concentration, and rise duration have been shown to predict alcohol-induced blackouts in college drinkers (Richards et al., 2024), and TAC signal processing models have been developed and validated against self-reported and breath alcohol concentration data in similar populations (Kianersi et al., 2023). Our model uses ideas from...—insert literature, ie michaelis menten— ...to characterize—the big picture formulation ie between drinking time, within episode duration, within episode drinking patterns—as a dynamical system. Compared to previous models, ours provide interpretable parameters with mechanistic meanings. Furthermore, we are able to distinguish between the end of a drinking episode and a return to baseline TAC. We show that the parameters can be recovered with typical TAC sample sizes.

simulation demonstrate the accuracy of our method/software.

We illustrate how these parameters could be used to characterize the effects of risk factors on binge-drinking in college students.

the remainder of this paper is organized as follows: —big long run-on sentence—

the criteria for a good model

We want the theory to achieve the following things: 1. It should operationalized by mechanistic parameters that are directly actionable by interventions.

2. It should be able to recreate things like disproportionate next day drinking consequences between males and females given similar rates. (ie AUC should be higher for people with lower metabolisms). If there have been disproportionate predictive validity in things like rise rate/fall rate/peak for predicting next day consequences based on sex, those same discontinuities should be accounted for by the metabolism difference.

3. It should be able to recreate the TAC "tail" behavior (is it linear or exponential?).

the theoretical model

While previous work has focused on mathematical modeling of between-person population dynamics (Duffy & Alanko, 1992; Gruenewald & Nephew, 1994), we focus on creating a within-person model of binge-drinking.

We want the model to capture the rate of entering an episode when in a non-drinking episode, the rate of exiting an episode when in a drinking episode, the within-episode dynamics, and the metabolism of alcohol.

Let $S(t) \in [0, 1]$ be the state of the individual at time t, where $S(t) = 0$ indicates a non-drinking episode and $S(t) = 1$ indicates a drinking episode. Our theory deconstructs binge-drinking into four components: the rate of entering a drinking episode, the rate of leaving a drinking episode, the within-episode ingestion rate, and the ethanol metabolism rate.

This could be formalized by,

$$TAC(t)dt = S(t)I(t) - M(t). \tag{1}$$

Where $I(t)$ represents the alcohol ingestion rate at time t and $M(t)$ represents the metabolic rate at time t. S(t) acts as a switching function, turning the intake rate function on or off depending on whether the individual is in a drinking episode or not. This is similar to previous work by (Duffy & Alanko, 1992) in that we model the time between drinking episodes as exponential and the number of drinks within each episode as Poisson distributed. We extend this work by explicitly relating these distributions to TAC at time t through a two-state continuous Markov process to characterize the between-episode dynamics and use of shot-noise process to characterize the within-episode dynamics. The switching function $S(t)$ is a two-state continuous Markov process with the transition rates q_{01} and q_{10}, representing the rates of entering and exiting a drinking episode, respectively.

There are a few options for modeling $I(t) + M(t)$. First is the model proposed by (Giraldo et al., 2017), which used a dampened spring oscillator to model the return to equilibrium

$$TAC(t)d^2t = -k^2TAC(t) - 2\zeta k\frac{dTAC(t)}{dt} + k^2\eta I(t), \tag{2}$$

where $k > 0$ is the natural frequency of the oscillator, $\zeta > 1$ is the damping ratio, and $\eta > 0$ is a scaling factor for the input function $I(t)$. One straightforward extension would be to model add a regime-switching component, so it handles the episodic nature of drinking.

Simulation Method

Describe the design of the simulation study: the data-generating model, the conditions manipulated, and the number of replications per condition.

Data-Generating Model

Specify the model used to generate the simulated data, including all fixed parameter values.

Design and Conditions

Describe the factors crossed in the simulation design (e.g., sample size, number of measurement occasions, effect size) and their levels.

Estimation and Evaluation Criteria

Describe the estimator(s) fit to each replication, the software used, and the criteria by which performance is evaluated (e.g., relative bias, RMSE, confidence interval coverage, convergence rates).

Simulation Results

Report the results of the simulation study, using APA-style statistical reporting throughout. Convergence rates and performance measures appear in Table 1, and Figure 1 shows how these measures vary with sample size and effect size.

Empirical Methods

Participants

Describe your sample: size, demographics, recruitment method, and any inclusion/exclusion criteria.

Materials

Describe instruments, apparatus, or materials used.

Procedure

Describe the step-by-step procedure of the study.

Data Analysis

Describe your analytic strategy (e.g., statistical tests, software used).

Empirical Results

Report your findings. Use APA-style statistical reporting, e.g., a significant effect was found, $t(28) = 3.45$, $p < .001$, $d = 0.65$. Regression estimates appear in Table 2, and Figure 2 shows the pattern of results.

Discussion

What were your main aims? What were key findings regarding each aim?

Compare and contrast findings observed to past literature.

Limitations and Future Directions

At least 2 limitations. How do your findings contribute to existing literature?

How should we move forward in research or practice given your contribution to the literature?

Conclusion

Summarize the key takeaways of your paper.

References

Duffy, J. C., & Alanko, T. (1992). Self-Reported Consumption Measures in Sample Surveys: A Simulation Study of Alcohol Consumption. *Journal of Official Statistics*, *8*(3), 327. https://ezaccess.libraries.psu.edu/login?url=https://www.proquest.com/scholarly-journals/self-reported-consumption-measures-sample-surveys/docview/1266814367/se-2?accountid=13158

Fairbairn, C. E., Han, J., Caumiant, E. P., Benjamin, A. S., & Bosch, N. (2025). A wearable alcohol biosensor: Exploring the accuracy of transdermal drinking detection. *Drug and Alcohol Dependence*, *266*, 112519. https://doi.org/10.1016/j.drugalcdep.2024.112519

Giraldo, L. F., Passino, K. M., & Clapp, J. D. (2017). Modeling and Analysis of Group Dynamics in Alcohol-Consumption Environments. *IEEE Transactions on Cybernetics*, *47*(1), 165–176. https://doi.org/10.1109/TCYB.2015.2509444

Greenfield, T. K. (2000). Ways of measuring drinking patterns and the difference they make: Experience with graduated frequencies. *Journal of Substance Abuse*, *12*(1–2), 33–49. https://doi.org/10.1016/S0899-3289(00)00039-0

Gruenewald, P. J., & Nephew, T. (1994). Drinking in California: Theoretical and empirical analyses of alcohol consumption patterns. *Addiction*, *89*(6), 707–723. https://doi.org/10.1111/j.1360-0443.1994.tb00957.x

Gunn, R. L., Merrill, J. E., Haines, A. M., Fernandez, M. E., Souza, T., Berey, B. L., Leeman, R. F., Wang, Y., & Barnett, N. P. (2023). Use of the BACtrack Skyn alcohol biosensor: Practical applications for data collection and analysis. *Addiction*, *118*(8), 1586–1595. https://doi.org/10.1111/add.16207

Kianersi, S., Ludema, C., Agley, J., Ahn, Y.-Y., Parker, M., Ideker, S., & Rosenberg, M. (2023). Development and validation of a model for measuring alcohol consumption from transdermal alcohol content data among college students. *Addiction*, *118*(10), 2014–2025. https://doi.org/10.1111/add.16228

Richards, V. L., Glenn, S. D., Turrisi, R. J., Mallett, K. A., Ackerman, S., & Russell, M. A. (2024). Transdermal alcohol concentration features predict alcohol-induced blackouts in college students. *Alcohol, Clinical and Experimental Research*, *48*(5), 880–888. https://doi.org/10.1111/acer.15290

Table 1

Simulation Performance Measures

N	True β_1	Bias	RMSE	Coverage	Power	Convergence
50	0.00	0.011	0.150	0.942	0.058	1.000
50	0.20	0.011	0.150	0.942	0.320	1.000
50	0.50	0.011	0.150	0.942	0.920	1.000
100	0.00	-0.000	0.103	0.956	0.044	1.000
100	0.20	-0.000	0.103	0.956	0.508	1.000
100	0.50	-0.000	0.103	0.956	1.000	1.000
250	0.00	-0.000	0.066	0.954	0.046	1.000
250	0.20	-0.000	0.066	0.954	0.868	1.000
250	0.50	-0.000	0.066	0.954	1.000	1.000

Note. Bias, RMSE, and 95% CI coverage/power are computed over converged replications only.

Convergence is the proportion of replications that converged.

Table 2

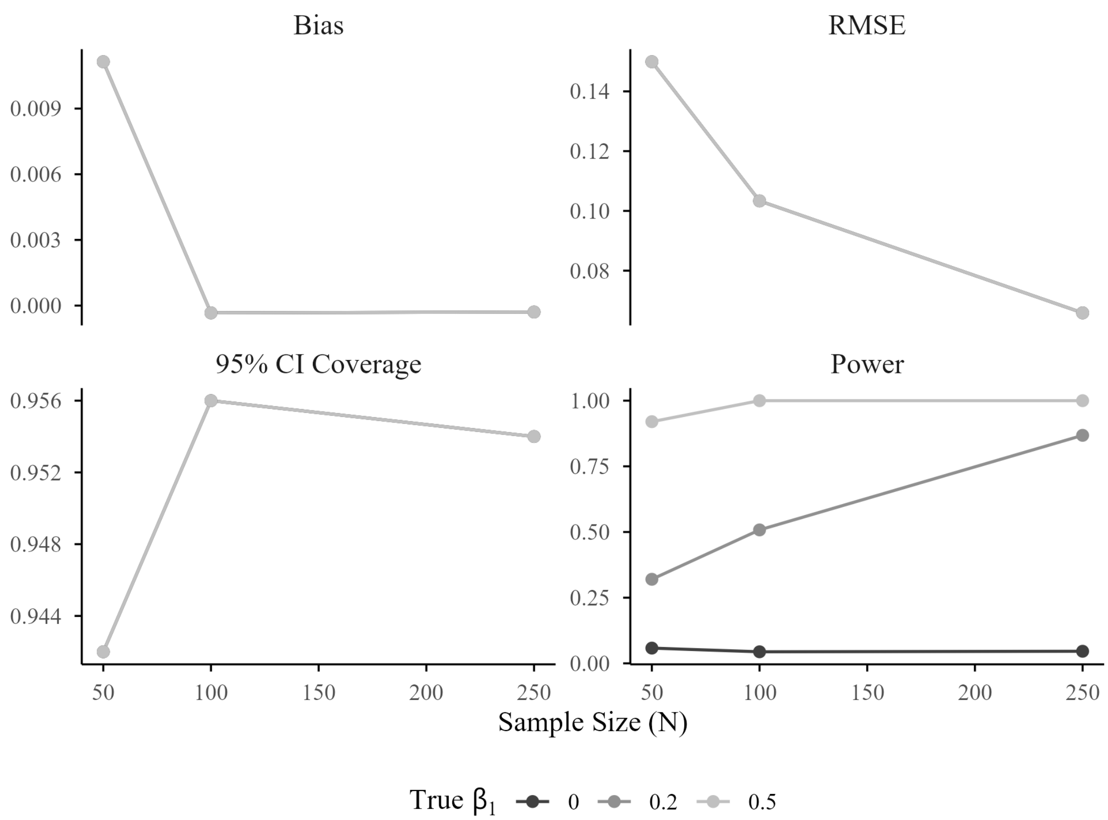
Empirical Analysis: Regression Estimates

Term	B	SE	p	95% CI
Intercept	7.721	3.162	0.015	[1.491, 13.951]
X_1	0.369	0.060	0.000	[0.250, 0.487]
Treatment	2.133	1.122	0.059	[-0.077, 4.344]

Note. N = 233, R^2 = 0.154.

Figure 1

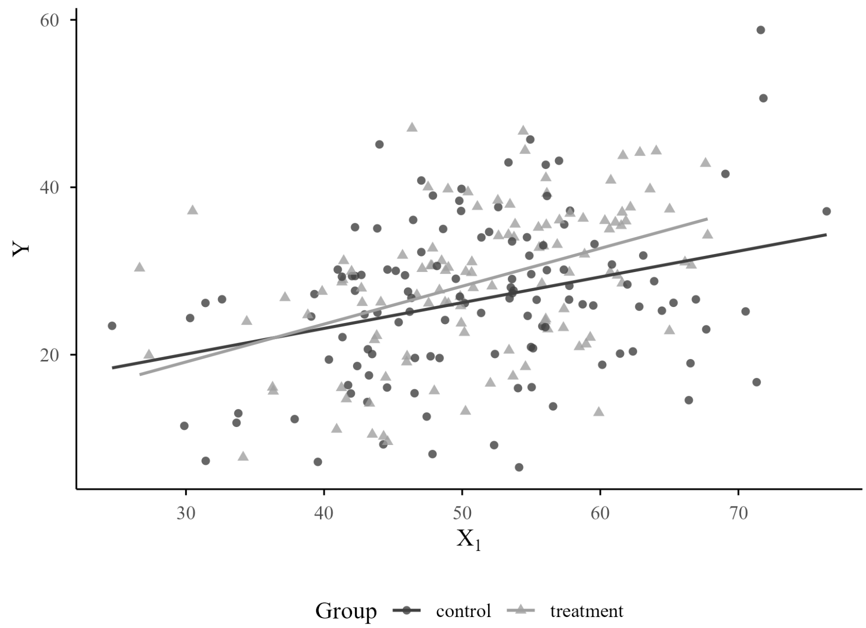
Simulation Performance Measures by Sample Size and Effect Size



Note. Bias, RMSE, and 95% CI coverage/power are computed over converged replications only.

Figure 2

Outcome by Predictor and Group



Note. Points are individual observations; lines are group-wise fitted values.